

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

```
df = pd.read_csv("/content/customer_age_spending_score.csv")
```

```
df.head()
```

	Customer_ID	Age	Spending_Score
0	1	19	39
1	2	21	81
2	3	20	6
3	4	23	77
4	5	31	40

```
df.tail()
```

	Customer_ID	Age	Spending_Score
15	16	32	55
16	17	41	35
17	18	29	70
18	19	38	25
19	20	24	88

```
X = df[["Age", "Spending_Score"]]
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

